

Telecom Network Intelligence System

1 System Overview

The Telecom Network Intelligence System (TNIS) is a telecom analytics platform designed to monitor network activity and identify congestion patterns. The system processes Call Detail Records (CDR), SMS logs, and internet usage data to generate analytical insights and congestion predictions.

Main objectives:

- Detect peak traffic regions
- Analyze usage trends
- Predict congestion risk
- Provide REST APIs and dashboard analytics

2 Architecture and Data Flow

The system follows a layered pipeline:

1. Raw CSV files are collected from telecom towers
2. Files are validated using Airflow
3. PySpark cleans and processes the data
4. Processed data is stored in Parquet format
5. Warehouse tables are loaded
6. FastAPI exposes analytical endpoints
7. React dashboard visualizes KPIs

2.1 Key Transformations

Stage	Transformation
Raw to Cleaned	Remove nulls, fix timestamps, rename columns
Cleaned to Enriched	Add region metadata
Enriched to Aggregated	Compute hourly and regional KPIs
Aggregated to Warehouse	Insert into fact and dimension tables

3 Warehouse Schema

The warehouse uses a star schema with:

- fact_usage
- dim_time
- dim_region

3.1 Sample Query

```
SELECT r.region_name ,
       t.hour ,
       SUM(f.call_count) AS total_calls
FROM fact_usage f
JOIN dim_time t ON f.time_id = t.time_id
JOIN dim_region r ON f.region_id = r.region_id
GROUP BY r.region_name , t.hour
ORDER BY total_calls DESC
LIMIT 10;
```

4 Batch vs Streaming

Data Flow	Mode	Reason
Network activity logs	Streaming	Requires near real-time monitoring
Daily usage reports	Batch	Generated periodically
Billing reports	Batch	Monthly processing is sufficient
Congestion alerts	Streaming	Requires immediate detection
Dashboard analytics	Batch	Hourly or daily refresh is acceptable

5 Storage Strategy

Layer	Format	Purpose
Raw Layer	CSV	Original source storage
Processed Layer	Parquet	Efficient analytics processing
Warehouse	PostgreSQL	API and reporting queries
ML Features	CSV/Parquet	Model training data

6 Machine Learning

The ML model predicts congestion risk levels:

- LOW
- MEDIUM
- HIGH

Features used:

- Average usage

- Growth rate
- Peak ratio
- Usage variability

Model used:

- Random Forest Classifier

Evaluation metrics:

- Accuracy
- Precision
- Recall
- F1-score

7 API Endpoints

Endpoint	Method	Purpose
/usage/summary	GET	Usage statistics
/usage/region/{region}	GET	Regional trends
/usage/peak	GET	Peak traffic details
/predict-usage-risk	POST	Congestion prediction

8 Technology Stack

Component	Technology
Data Processing	Python, Pandas
Distributed ETL	PySpark
Orchestration	Apache Airflow
Warehouse	PostgreSQL
Backend API	FastAPI
Frontend	React
Machine Learning	scikit-learn
Storage Format	Parquet